DEEP-MIRT: Dynamic Psychometric Knowledge Tracing for Partial-Credit and Rubric-Scored Assessment

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Abstract

Knowledge tracing is a core component of adaptive learning systems, yet much of the literature assumes binary correctness as the observation signal. In many educational settings, however, learner evidence is polytomous, including ordered rubric levels and partial credit, where both ordinal structure and interpretability matter for measurement and intervention.

We present DEEP-MIRT, a modular framework that extends knowledge tracing to polytomous responses by bridging sequential neural modeling with psychometric Item Response Theory. The framework couples a swappable sequential backbone with an interchangeable IRT-grounded response head, separating the question of how learner ability evolves over time from the question of how that ability maps to ordered response categories. Instantiated with a Generalized Partial Credit Model (GPCM) head, DEEP-MIRT produces interpretable outputs at every interaction: an evolving student ability estimate, per-item discrimination values, and ordered difficulty thresholds corresponding to rubric level transitions. On synthetic benchmarks with known ground truth, DEEP-GPCM achieves strong rank-order agreement across category resolutions $K \in \{2, 3, 4, 5\}$ (QWK up to 0.778) and recovers item difficulty thresholds and discrimination with high to moderate fidelity (mean $r_\beta = 0.90$; $r_\alpha \approx 0.65$). The model remains backward-compatible with standard binary knowledge tracing benchmarks, matching established baselines on ASSIST2015 and Synthetic-5 with $K = 2$. Together, the proposed model advances knowledge tracing from a binary predictive instrument into a measurement-aware framework that supports both instructional decision-making with psychometric interpretation.

Introduction

Educational assessments often target graded performance rather than pass/fail outcomes. Rubric-based scoring, partial-credit items, and constructed-response tasks encode degrees of correctness that binary labels collapse into a single signal (**??**). While automated scoring systems can deliver this ordinal structure at scale in intelligent tutoring contexts (**??**), most KT benchmarks, evaluation protocols, and dominant modeling

paradigms still assume binary correctness. This leaves ordinal outcomes comparatively underexplored in KT, even though they are central to authentic assessment practice.

Beyond the outcome type, interpretability remains a key concern. Most KT models learn high-dimensional states optimized for prediction, but these states are not, by construction, tied to interpretable quantities. Therefore, constructs that educators and assessment designers routinely work with, such as student ability level, item discriminatory power, or the difficulty steps between adjacent rubric categories, are not directly identifiable. Item Response Theory (IRT) provides exactly this structure through identifiable parameters including student ability, item discrimination, and ordered step difficulties, grounding score interpretation in established psychometric theory (**???**). Standard IRT models, however, treat ability as fixed and are not designed for sequential interaction data. This work therefore extends KT to polytomous responses by incorporating the interpretable parameter structure of IRT within a dynamic sequential framework.

Three lines of existing work address these issues in isolation, leaving a gap at their intersection. First, deep KT models, from Deep Knowledge Tracing (DKT; **?**) to memory-augmented and attention-based variants (**??**), capture temporal learning dynamics but typically assume dichotomous correctness and learn latent states with limited psychometric interpretability. Second, polytomous Item Response Theory (IRT) models such as the Generalized Partial Credit Model (GPCM; **?**) provide principled, interpretable measurement of ordered responses with identifiable parameters (student ability, item discrimination, ordered step thresholds), but they generally treat ability as time-invariant and are not designed for sequential interaction logs. Third, recent efforts that inject IRT structure into deep KT (**??**) improve interpretability but largely remain restricted to dichotomous outcomes. An early partial exception is **?**, who extended Bayesian Knowledge Tracing to allow partial credit via continuous Bayesian nodes, but without deep sequential modeling or recovering standard polytomous IRT step structure. Yet these approaches has not yet reconciled sequential dynamics with ordinal response modeling while retraining psychometric identifiability.

We propose *DEEP-MIRT*, a modular framework that extends KT to Polytomous Knowledge Tracing (PKT) by coupling a swappable sequential backbone with an IRT-grounded response head. In this setting, PKT is the task of predicting the full distribution over ordered response categories from a learner's interaction history. We instantiate the framework with a GPCM response head, yielding DEEP-GPCM, which combines a Dynamic Key-Value Memory Network (DKVMN) backbone with a GPCM observation model. At each timestep, the GPCM head maps the learner state to a distribution over ordinal categories and exposes psychometrically interpretable components, including an evolving student ability estimate, as well as learned item parameters such as per-item discrimination values and ordered step difficulties corresponding to rubric level transitions. While the experiments focus on ordinal PKT, the framework accommodates other polytomous formats through alternative response heads.

Our main contributions are:

1. **A modular framework for polytomous knowledge tracing.** We formulate PKT as predicting the full ordinal response distribution from interaction history and propose DEEP-MIRT, which decouples a sequential backbone (encoder, mem-

ory, summary) from an interchangeable psychometric response head. The head interface accepts student ability θ_t and item identity $\mathbf{k}_t$ and returns a category probability vector; any differentiable observation model satisfying this contract is a valid instantiation. This design separates the educational measurement question (which model governs how ability maps to rubric scores?) from the knowledge tracing question (how does ability evolve over time?).

2. **GPCM instantiation with interpretable psychometric parameters.** We instantiate DEEP-MIRT with a GPCM-based response head as DEEP-GPCM, incorporating a monotonic threshold parameterization that enforces rubric ordering by construction. The resulting model produces evolving ability trajectories, item discrimination estimates, and ordered step difficulty thresholds, all of which correspond to constructs that educators and assessment designers already use in rubric design and standard-setting practice.

3. **Scalable item encoding and empirical evaluation.** We propose Static Item Embedding (SIE) for large item banks and evaluate ordinal prediction quality, psychometric parameter recovery, ablation of design components, and backward compatibility with binary KT benchmarks.

We investigate the following research questions:

- **RQ1 (Prediction).** Does coupling a DKVMN backbone with an ordinal, psychometrically-constrained GPCM response head improve polytomous next-response prediction, and does it better preserve ordinal structure than format-unaware alternatives?

- **RQ2 (Parameter Recovery).** Do the IRT parameters inferred by DEEP-GPCM faithfully recover known ground-truth values from synthetic data, and are item parameter estimates stable across student subpopulations as assessed by split-half reliability?

- **RQ3 (Learner State).** Do the inferred student ability trajectories θ_t exhibit psychologically plausible learning dynamics, and can DEEP-GPCM reveal meaningful learner patterns (mastery, growth, persistent difficulty) that are not visible from format-unaware latent state representations?

- **RQ4 (Scalability).** Does Static Item Embedding maintain ordinal prediction quality and parameter recovery at large item-bank scales ($Q \sim 5{,}000$) while substantially reducing parameter count compared to dense encoding strategies?

- **RQ5 (Ecological Validity).** When ordinal response categories are constructed from proxy signals in existing binary KT benchmarks (e.g., attempt counts), does the ordinal GPCM head yield measurable gains over format-unaware alternatives, and how sensitive are these gains to the quality of the proxy ordinal construction?

Related Work

Knowledge Tracing: From Binary to Polytomous

Knowledge tracing has evolved through several modeling paradigms, but all share a common assumption: learner responses are binary. Bayesian Knowledge Tracing (BKT) models mastery of a Knowledge Component (KC) as a two-state Hidden Markov Model (**?**). Deep Knowledge Tracing (DKT) reframed the task as sequence modeling with recurrent neural networks, learning high-dimensional latent states from interaction sequences (**?**). Dynamic Key-Value Memory Networks (DKVMN) introduced a structured memory that separates static concept representations (key memory) from evolving mastery states (value memory), offering a more transparent mapping between knowledge components and latent state than generic RNNs (**??**). Attention-based models such as SAKT and AKT apply self-attention to weigh the relevance of prior interactions for next-step prediction (**??**), and recent work has further explored Transformer-based architectures (**??**). Throughout this progression, the response format has remained dichotomous: every model cited above predicts a binary correct/incorrect outcome.

Recent KT research has begun to acknowledge that correctness alone is insufficient. **?** fuse process features (attempts, response time, historical accuracy) to enrich difficulty representations, and **?** integrate problem complexity and state-stability mechanisms to reduce noisy updates. These efforts treat richer signals as auxiliary features within a binary prediction framework. They do not change the response format itself, but they highlight that educational interactions carry information beyond pass/fail: precisely the information that ordinal and polytomous responses encode directly. The only prior work to move beyond binary responses in a KT context is **?**, who extended BKT to partial credit using continuous Bayesian nodes, but without deep sequential modeling or IRT-based parameterization.

Interpretability in Deep Knowledge Tracing

In most DKT models, learned states are opaque distributed representations rather than psychometrically meaningful constructs (**?**). Several efforts address this gap by coupling neural architectures with Item Response Theory (IRT) structure. Deep-IRT attaches an IRT-style readout to DKVMN, exposing student ability and item difficulty parameters for each prediction (**?**). DKVMN-MRI similarly integrates IRT within an LSTM-based memory framework (**?**). **?** propose a general approach for incorporating IRT parameters into KT training. These IRT-augmented models improve interpretability: a teacher or assessment specialist can inspect the estimated ability and difficulty values, rather than relying on opaque hidden states. In educational terms, "interpretability" here means that model outputs correspond to constructs that practitioners already use (student proficiency levels, item difficulty, and discrimination quality), supporting the kinds of validity arguments that responsible assessment requires (**??**).

A critical limitation, however, is that all existing IRT-KT integrations operate on dichotomous responses. They estimate a single ability and difficulty for binary outcomes and do not model the ordered step structure of rubric scores or partial-credit items.

Attention-based models provide a complementary form of interpretability through attention weights (which interactions were most relevant to a prediction), but they too lack psychometric parameterization and ordinal response awareness.

Polytomous Item-Response Theory

Psychometrics provides a family of models for ordered responses grounded in a latent ability parameter θ and item-specific properties (**??**). Two traditions dominate for ordinal outcomes. Cumulative models such as the Graded Response Model (GRM; **?**) define boundary parameters between adjacent categories, modelling the probability of scoring at or above category k as a logistic function of θ. Adjacent-category models, including the Partial Credit Model (PCM; **?**) and the Generalized Partial Credit Model (GPCM; **?**), parameterize each step transition directly: the GPCM characterizes each item by a discrimination parameter α_j controlling how sharply the item differentiates ability levels, and ordered step difficulty thresholds $\boldsymbol{\beta}_j = (\beta_{j,0}, \ldots, \beta_{j,K-2})$ representing the ability level at which each consecutive rubric transition becomes equally likely. Adjacent-category models are the natural fit for rubric scoring, where performance progresses sequentially through ordered levels (**??**). For unordered categories, the Nominal Response Model (NRM; **?**) provides an analogous framework.

All classical polytomous IRT models treat ability as fixed across items, which is appropriate for standardized testing but inadequate for adaptive tutoring contexts where ability evolves through practice. Extending the interpretable parameter structure of IRT to a dynamic sequential setting is the central goal of this work.

Methodology

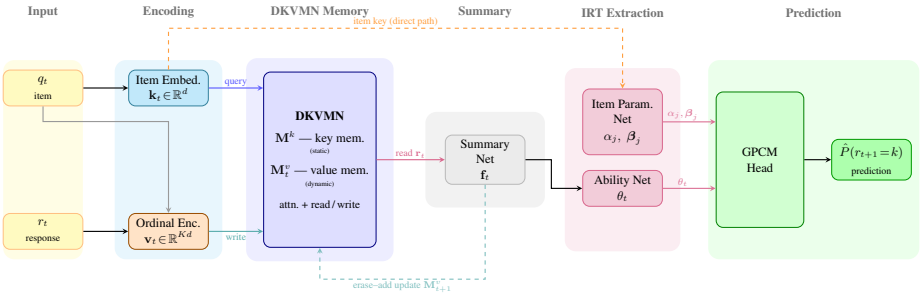

Figure 1: DEEP-MIRT framework at timestep t. Item parameters $(\alpha_j, \boldsymbol{\beta}_j)$ bypass the student state via the item embedding (dashed, top), consistent with the IRT assumption that discrimination and difficulty are student-independent. The GPCM head maps $(\theta_t, \alpha_j, \boldsymbol{\beta}_j)$ to the predicted ordinal distribution $\hat{P}(r_{t+1} = k)$.

Framework Overview

DEEP-MIRT is a modular architecture with two separable concerns. The *backbone* handles the knowledge tracing problem: it encodes item–response interactions, maintains a dynamic memory of the student's evolving knowledge state, and produces a summary vector $\mathbf{f}_t$ representing current student state. The *response head* handles the measurement problem: it maps the backbone's student state $\mathbf{f}_t$ and item identity $\mathbf{k}_t$ to a probability distribution over response categories through a psychometric observation model. Any differentiable function $h(\mathbf{f}_t, \mathbf{k}_t) \rightarrow \Delta^{K-1}$ (a distribution over K categories) satisfying this interface is a valid DEEP-MIRT instantiation. The backbone is agnostic to which measurement model governs the head.

This separation is conceptually and practically helpful for tracking learning progressions. The backbone addresses the learning dynamics question of how student ability evolves across a sequence of interactions, naturally handled by sequential neural architectures. The head addresses the measurement question of how current ability maps to the probability of each ordered rubric score, naturally handled by psychometric theory. Coupling them through a clean interface means the two concerns can be improved independently: a better backbone (e.g., Transformer instead of DKVMN) improves temporal modeling without changing the interpretable parameterization; a different head (e.g., GRM instead of GPCM) changes the measurement assumptions without retraining the backbone.

In this paper we instantiate the head as the **Generalized Partial Credit Model** (GPCM, Section **??**). We chose GPCM over the other major polytomous families for three reasons: (1) it is an adjacent-category model, which is the natural match for rubric scoring where a student progresses sequentially through performance levels; (2) it produces a discrimination parameter α_j and ordered step difficulty thresholds $\boldsymbol{\beta}_j$ that correspond directly to rubric design decisions and can be inspected by assessment practitioners; (3) its monotone threshold constraint admits a clean differentiable parameterization (Section **??**). A cumulative model such as the Graded Response Model (**?**) would be a natural alternative for tasks where each scoring threshold represents an independent decision (e.g., analytic rubrics with independent sub-scores); a Nominal Response Model (**?**) would apply when response categories are unordered (e.g., multiple-choice distractors encoding different misconceptions). We treat these as directions for future instantiation.

The backbone processes inputs through four sequential stages as shown in Figure **??**: ordinal interaction encoding, DKVMN memory update, summary compression, and IRT-based parameter extraction. An ordinal-aware training objective supervises all stages jointly. The following subsections formalize each component.

Problem Formulation

Let $r_t \in \{0, 1, \ldots, K-1\}$ denote an observed response with K *ordered* categories (e.g., rubric levels / partial credit), and let $q_t \in \{0,1\}^Q$ be a one-hot indicator of the exercised question/item, where Q is the number of unique items. An interaction is $x_t = (q_t, r_t)$ and the observed history up to time t is $\mathcal{X}_t = \{(q_1, r_1), \ldots, (q_t, r_t)\}$. We

study *ordinal* (polytomous) knowledge tracing as predicting the full distribution

$$P(r_{t+1} = k \mid q_{t+1}, \mathcal{X}_t), \quad k \in \{0, \ldots, K-1\}. \tag{1}$$

Ordinal-Aware Interaction Representation

Linear Decay Encoding We encode an ordinal response r_t with a "triangular" kernel over ordered categories. Define a length-K ordinal code $\mathbf{s}_t \in \mathbb{R}^K$ by

$$\begin{aligned} s_t^{(k)} &= \max\left(0, 1 - \frac{|k - r_t|}{K-1}\right), \quad 0 \le k \le K-1, \\ \mathbf{s}_t &= [s_t^{(0)}; \ldots; s_t^{(K-1)}]. \end{aligned} \tag{2}$$

For example, with a $K = 4$ rubric and a response of $r_t = 2$ ("correct approach with errors"), the ordinal code is $\mathbf{s}_t = [0.33,\ 0.67,\ 1.0,\ 0.67]$, which spreads activation to neighboring categories while peaking at the observed level. This encoding preserves the ordinal structure: a response of 2 is represented as more similar to responses 1 and 3 than to response 0, reflecting the educational intuition that partial credit is closer to full credit than to no credit.

We then bind the ordinal code to the item indicator via a Kronecker product (equivalently, a block one-hot expansion):

$$\mathbf{x}_t = q_t \otimes \mathbf{s}_t \in \mathbb{R}^{KQ}. \tag{3}$$

This reduces to the standard binary interaction encoding when $K = 2$ (consistent with **?**). While this encoding is effective for moderately sized item banks, the $\mathcal{O}(KQ)$ dimensionality becomes a practical constraint for large Q; we address this in the following subsection.

Scalable Value Encoding for Large Item Banks Large-scale educational platforms maintain extensively large item banks with various grade levels. The triangular Kronecker encoding in (**??**) yields an interaction vector of dimension KQ, which grows linearly with item bank size. The subsequent value projection $\mathbf{v}_t = B^\top \mathbf{x}_t \in \mathbb{R}^{d_v}$ therefore requires a $KQ \times d_v$ weight matrix; at $Q = 5{,}000$ and $K = 5$, this amounts to 1.6M parameters and a per-batch intermediate tensor of size $\mathrm{batch} \times S \times 25{,}000$, placing substantial memory pressure on GPU-constrained deployment environments.

We propose **Static Item Embedding** (SIE), an alternative value encoding that decouples item identity from response pattern via two independent additive branches:

$$\mathbf{v}_t = W_{\text{item}}\, \mathbf{u}_{q_t} + W_{\text{resp}}\, \mathbf{s}_t + \mathbf{b}_{\text{resp}} \in \mathbb{R}^{d_v}, \tag{4}$$

where $\mathbf{u}_{q_t} \in \mathbb{R}^H$ is a *frozen* (non-trainable) unit-norm random vector assigned to item q_t at initialization, $\mathbf{s}_t \in \mathbb{R}^K$ is the triangular ordinal code from (**??**), and $W_{\text{item}} \in \mathbb{R}^{d_v \times H}$ and $W_{\text{resp}} \in \mathbb{R}^{d_v \times K}$ are learned projections shared across all items.

The factored structure in (**??**) separates two gradient channels: W_{item} is updated by item-identity signals, while W_{resp} is updated by response-pattern signals. Because item vectors are frozen, items cannot collapse to similar representations during training, a

failure mode of learned separable embeddings at large Q (cf. Section **??**). For two independent uniformly random unit vectors in $\mathbb{R}^H$, the expected squared inner product is $1/H$ (**?**), so pairwise interference concentrates around zero as H grows; cross-item gradient crosstalk in W_{item} therefore decays with H independently of Q, making SIE well-suited for large item banks.

The frozen dimension H is set as the smallest power of two no less than $\max(128, \lfloor Q/2 \rfloor)$, clamped to $[128, 1024]$.[1] This schedule ensures near-orthogonality scales with item bank size while keeping the projection dimension bounded, in contrast to the $\mathcal{O}(KQ)$ complexity of the full Kronecker encoding.

Dynamic Key-Value Memory Network

We adopt the Dynamic Key-Value Memory Network (DKVMN) as the sequential backbone. Let $\mathbf{k}_t = A^\top q_t \in \mathbb{R}^{d_k}$ be a key embedding for the current item and $\mathbf{v}_t \in \mathbb{R}^{d_v}$ be a value embedding for the interaction, obtained via the Kronecker projection $\mathbf{v}_t = B^\top \mathbf{x}_t$ (ordinal-aware encoding) or the Static Item Embedding in (**??**) (scalable encoding). The model maintains a static key memory $\mathbf{M}^k \in \mathbb{R}^{N \times d_k}$ and a dynamic value memory $\mathbf{M}_t^v \in \mathbb{R}^{N \times d_v}$. Educationally, the key memory represents latent skill areas or knowledge components, the value memory tracks the student's evolving mastery of each area, and the attention weights $\mathbf{w}_t$ (below) indicate which skill areas a given item exercises. The summary vector $\mathbf{f}_t$ aggregates the student's current mastery state relevant to the item at hand.

Specifically, the key query is first projected via a learned transform, and attention weights over memory slots are computed as

$$\tilde{\mathbf{k}}_t = \tanh(W_{\text{attn}} \mathbf{k}_t), \qquad \mathbf{w}_t = \text{softmax}(\mathbf{M}^k \tilde{\mathbf{k}}_t) \in \mathbb{R}^N, \qquad w_t(i) = [\mathbf{w}_t]_i, \quad (5)$$

and the read vector is

$$\mathbf{r}_t = \sum_{i=1}^{N} w_t(i)\, \mathbf{M}_t^v(i) \in \mathbb{R}^{d_v}. \quad (6)$$

We fuse item context and retrieved state into a summary vector

$$\mathbf{f}_t = \tanh(W_{\mathbf{f}}[\mathbf{r}_t; \mathbf{k}_t] + b_{\mathbf{f}})\,, \quad (7)$$

which is then used by the response head to predict r_{t+1}.

Following memory-augmented neural networks, the value memory is updated via erase-add operations:

$$\mathbf{e}_t = \sigma(W_{\mathbf{e}} \mathbf{v}_t + b_{\mathbf{e}}), \quad (8)$$

$$\mathbf{a}_t = \tanh(W_{\mathbf{a}} \mathbf{v}_t + b_{\mathbf{a}}), \quad (9)$$

$$\mathbf{M}_{t+1}^v(i) = \mathbf{M}_t^v(i)\big[1 - w_t(i)\mathbf{e}_t\big] + w_t(i)\mathbf{a}_t. \quad (10)$$

[1] Concretely, $H = 128$ for $Q \leq 256$, $H = 256$ for $Q \approx 500$, $H = 512$ for $Q \approx 1{,}000$, and $H = 1{,}024$ for $Q \geq 2{,}000$.

DEEP-GPCM: an Ordinal IRT Head

Instead of the binary readout (e.g., $\sigma(W_p^\top \mathbf{f}_t + b_p)$), we use a Generalized Partial Credit Model (GPCM) head to obtain category probabilities for ordered responses. Let j denote the item exercised at time t (i.e., $[q_t]_j = 1$). We compute a time-varying ability proxy θ_t and item-specific parameters (thresholds and discrimination) as

$$\theta_t = W_\theta \mathbf{f}_t + b_\theta, \tag{11}$$

$$\beta_{j,0} = W_\beta \mathbf{k}_t + b_\beta, \tag{12}$$

$$\beta_{j,k} = \beta_{j,k-1} + \mathrm{softplus}(W_{g[k]} \mathbf{k}_t + b_{g[k]}), \quad 1 \le k \le K-2, \tag{13}$$

$$\alpha_j = \exp(W_\alpha \mathbf{k}_t + b_\alpha)\,. \tag{14}$$

The construction in (**??**) enforces $\beta_{j,0} < \beta_{j,1} < \cdots < \beta_{j,K-2}$ by adding strictly positive increments (via softplus) to each preceding threshold. This monotonicity guarantee has direct educational meaning: achieving rubric level $k{+}1$ always requires higher ability than achieving level k, respecting the ordering that a rubric designer intended. Without this constraint, the model could learn disordered thresholds where some rubric levels are never the most probable response at any ability level, a well-documented problem in educational measurement that signals a poorly functioning item (**?**).

Because discrimination is an item-intrinsic property in the GPCM, α_j is derived solely from the item key $\mathbf{k}_t$, deliberately excluding the student-state summary $\mathbf{f}_t$. This separation ensures that the estimated discrimination reflects the item's inherent capacity to differentiate ability levels, independently of which student is currently being modelled — consistent with the IRT assumption that item parameters are population-invariant (**?**). Item-level discrimination is recovered by averaging per-timestep estimates across all encounters with a given item (**?**). The exponential activation ensures $\alpha_j > 0$.

Given $(\theta_t, \alpha_j, \{\beta_{j,k}\}_{k=0}^{K-2})$, the GPCM category probability is

$$p_{t,k} = \frac{\exp\Big(\sum_{i=0}^{k-1} \alpha_j(\theta_t - \beta_{j,i})\Big)}{\sum_{c=0}^{K-1} \exp\Big(\sum_{i=0}^{c-1} \alpha_j(\theta_t - \beta_{j,i})\Big)}. \tag{15}$$

At each timestep, the model evaluates this standard GPCM formulation (cf. Section **??**) using the *dynamically estimated* student ability θ_t and item parameters $\alpha_j, \boldsymbol{\beta}_j$. Unlike classical GPCM estimation where ability is static, here θ_t evolves as new interactions are observed, enabling the model to track learning over time while retaining the interpretable measurement structure of polytomous IRT.

We predict $\hat{r}_{t+1} = \arg\max_k p_{t,k}$, and optimize the combined ordinal loss in (**??**) over the observed responses.

Training Objective for Ordinal Data

Because the GPCM defines category probabilities via (**??**) rather than cumulative links, we optimize a weighted categorical objective with an ordinal distance penalty to mitigate class imbalance (**?**). The loss combines two complementary components:

$$\mathcal{L} = \lambda_{\text{focal}} \mathcal{L}_{\text{focal}} + \lambda_{\text{WOL}} \mathcal{L}_{\text{WOL}}, \tag{16}$$

where $\mathcal{L}_{\text{focal}}$ is the focal loss (**?**), which down-weights well-classified examples and focuses training on ambiguous responses near category boundaries. The weighted ordinal loss $\mathcal{L}_{\text{WOL}}$ is

$$\mathcal{L}_{\text{WOL}} = \frac{1}{n} \sum_{i=1}^{n} \Big[(1 + \gamma |y_i - \hat{y}_i|) \cdot w_{y_i} \cdot (-\log \hat{p}_{i,y_i}) \Big], \tag{17}$$

where y_i is the true category, $\hat{p}_{i,y_i}$ is the predicted probability under (**??**), $\hat{y}_i = \arg\max_k \hat{p}_{i,k}$ is the predicted label, and γ controls the ordinal distance penalty. To balance classes, with c_k the sample count of class k, we set $w_k = \sqrt{\frac{n}{K \cdot c_k}}$, where n is the number of valid tokens in the batch.

Experiments

Datasets

Synthetic-Ordinal. To the best of our knowledge, there are no standardized public benchmarks for ordinal knowledge tracing. We therefore generate a synthetic dataset using an IRT-GPCM data generating process following established psychometric principles. This design provides controlled polytomous supervision $r_t \in \{0, \ldots, K-1\}$ and enables parameter recovery analysis with known ground truth.

We simulate $5{,}000$ students interacting with questions drawn from a pool of $Q = 200$ questions in random order. Each interaction sequence has variable length in $[50, 200]$. Students and questions are configured with *static* GPCM parameters following assumed priors:

- *Student ability:* $\theta \sim \mathcal{N}(0, 1)$,
- *Item discrimination:* $\alpha \sim \text{LogNormal}(0, 0.3)$,
- *Item thresholds:* $K-1$ values drawn independently from $\mathcal{N}(0, 1)$ and sorted to obtain $\beta_0 < \cdots < \beta_{K-2}$.

We generate four datasets with $K \in \{2, 3, 4, 5\}$ to test extensibility across category resolutions. Each dataset is split 80/20 for training/testing; a validation split is held out from training for early stopping and hyperparameter tuning.

Proxy-Ordinality To test our model on real educational logs, we derive ordinal labels from attempt counts in ASSISTments-style data (e.g., 1 attempt = mastery, 2–3 = partial, 4+ = struggle). Results and evaluation details are described in Section **??**.

Binary Compatibility After validating the PKT instantiations on Synthetic-Ordinal, we confirm compatibility with binary KT. Due to implementation differences in batch handling and sequence-to-question lookup, we select two datasets:

- **ASSIST2015**[2]: 19,840 students and 100 KCs/questions.

[2]https://sites.google.com/site/assistmentsdata/datasets/2015-assistments-skill-builder-data

- **Synthetic-5**[3]: (**?**) simulated 4,000 students' responses with probability $p(r) = c + \frac{1}{1+\exp(-(\theta-\beta))}$ where $c = 0.25$ models question guessing. Each student is administered 50 questions in sequential order.

Models and Baselines

We compare DEEP-GPCM against three baselines that progressively relax psychometric and ordinal structure, each isolating a different design question:

- **DKVMN + Softmax**: DKVMN backbone with a K-way softmax head trained with categorical cross-entropy. No ordinal structure, no IRT constraints. Tests whether psychometric parameterization provides gains over a purely neural polytomous baseline.
- **DKVMN + Ordinal**: DKVMN backbone with a linear output head trained using the weighted ordinal loss (**??**). Introduces ordinal awareness without IRT parameterization, isolating the contribution of the GPCM structure over simpler ordinal objectives.
- **Static GPCM**: Standard GPCM fit from interaction data with student ability held fixed across items. No sequential dynamics. Represents the psychometric status quo and quantifies what sequential modeling adds beyond classical IRT estimation.
- **Dynamic GPCM**: GPCM with per-item α/β lookup tables and a gated recurrent ability update driven by prediction error at each step. No DKVMN memory. Isolates the contribution of sequential ability updating from DKVMN's distributed memory.
- **DEEP-GPCM (ours)**: DKVMN backbone with GPCM head and combined ordinal loss (**??**). Uses SIE encoding by default; the choice of encoding strategy is ablated in Section **??**.

Prediction Mapping

For $K = 2$, predicted probabilities are converted to labels with a threshold of 0.5 following **?**. For $K > 2$, we use $\hat{y}_t = \arg\max_k \hat{p}_{t,k}$ to map categorical probabilities to prediction labels.

Training Setup

We use the combined loss in (**??**) to handle class imbalance and ordinal agreement, with tunable weights λ_*.

We set $d_k = 64$ and $d_v = 128$ for key/value embedding dimensions and configure $N = 50$ memory slots for both memory matrices.

All models are trained with batch size 64 using Adam with learning rate 10^{-3}, reduced on performance plateau as per **?** for late-epoch stability. We apply dropout

[3] https://github.com/chrispiech/DeepKnowledgeTracing/tree/master/data/synthetic

0.05 and early stopping if validation loss does not improve for 10 epochs. Models are implemented in PyTorch and trained on an RTX 4060 Laptop GPU.

Evaluation Metrics

We use the following metrics:

- **Categorical Accuracy (ACC)**: percentage of exact matches between predicted category and ground truth. While straightforward, ACC does not distinguish near-misses from far-misses and is therefore insufficient as a sole ordinal metric.
- **Quadratic Weighted Kappa (QWK)** (**?**): our primary metric for ordinal tasks. QWK applies a quadratic penalty proportional to the squared distance between predicted and true categories, making it the standard agreement measure in automated essay scoring and rubric evaluation. A QWK of 0.0 indicates chance-level agreement; 1.0 indicates perfect agreement.
- **Kendall's Tau (τ)**: measures rank-order agreement between predicted and true response sequences, penalizing all inversions equally regardless of distance.
- **Area Under the ROC Curve (AUC)**: used for the binary case ($K = 2$) to maintain comparability with prior KT literature.

IRT Analysis Setup

For parameter recovery analysis on Synthetic-Ordinal, we use:

- $\boldsymbol{\theta}$ **Aggregation**: we take the mean ability estimate $\bar{\theta} = \frac{1}{T}\sum_{t=1}^{T}\theta_t$ across all timesteps for each student, averaging out sequence-position noise while exploiting the full interaction history.
- $\boldsymbol{\alpha}, \boldsymbol{\beta}$ **Aggregation**: because questions appear with variable frequency due to random sequencing, we aggregate item parameters based on occurrences and use the sample mean for recovery analysis.

We normalize learned parameters and true parameters using the priors used to generate each dataset. Specifically, both θ and β use z-score normalization. α is log-transformed and standardized with log-mean and log-standard deviation; we then rescale by 0.3 before the final exponential projection to match the generating scale.

Results

Ordinal Prediction on Synthetic-Ordinal (RQ1)

Table **??** reports prediction performance on Synthetic-Ordinal for $K \in \{2, 3, 4, 5\}$, addressing **RQ1**.

Two findings stand out. First, the τ advantage of DEEP-GPCM over the DKVMN baselines is largest at $K = 3$ and shrinks as K grows, reaching near-parity at $K = 5$. This is counterintuitive if one attributes the gain to the GPCM's threshold structure —

Table 1: Prediction performance on Synthetic-Ordinal ($Q = 200$) across $K \in \{2, 3, 4, 5\}$. Best per column in bold.

(a) $K = 2$ (Binary)

Model	**Acc(%)**	**AUC**
Static GPCM	61.6	0.664
Dynamic GPCM	68.5	0.753
DKVMN + Softmax	70.33	0.776
DKVMN + Ordinal	70.59	0.778
DEEP-GPCM (ours)	**71.4**	**0.779**

(b) $K = 3$

Model	**Acc(%)**	**QWK**	τ
Static GPCM	48.5	0.313	0.233
Dynamic GPCM	57.2	0.555	0.407
DKVMN + Softmax	60.18	0.570	0.407
DKVMN + Ordinal	59.59	0.592	0.431
DEEP-GPCM (ours)	**60.7**	**0.618**	**0.567**

(c) $K = 4$

Model	**Acc(%)**	**QWK**	τ
Static GPCM	39.0	0.322	0.244
Dynamic GPCM	49.6	0.644	0.490
DKVMN + Softmax	53.58	0.650	0.482
DKVMN + Ordinal	52.58	0.684	0.516
DEEP-GPCM (ours)	**53.7**	**0.710**	**0.627**

(d) $K = 5$

Model	**Acc(%)**	**QWK**	τ
Static GPCM	37.4	0.390	0.298
Dynamic GPCM	45.3	0.717	0.553
DKVMN + Softmax	**51.73**	0.724	0.541
DKVMN + Ordinal	50.13	**0.762**	**0.582**
DEEP-GPCM (ours)	50.7	0.759	0.576

more thresholds should, if anything, help more at higher K. A more consistent explanation is that at low K, the ordinal signal per observation is weak and the GPCM's ability-centred parameterisation provides a stronger inductive bias for ranking students correctly; at high K, the richer per-step signal allows the DKVMN baselines to learn comparable rank structure from data alone. Second, the IRT ablation shows that sequential dynamics, not item parameterisation, drive the bulk of performance: Dynamic GPCM, which shares item tables with Static GPCM but adds a single gated ability update, recovers most of the gap to the neural models. At $K = 4$ and $K = 5$, Dynamic GPCM nearly matches DKVMN + Softmax on QWK despite having no distributed memory, which suggests that a lightweight recurrent IRT model is a surprisingly competitive baseline when the response scale is fine-grained enough to provide a strong per-step learning signal.

Binary Compatibility on Standard KT Benchmarks

An important practical requirement is that a polytomous KT model should not sacrifice performance when deployed on standard binary data, since many existing item banks and datasets use dichotomous scoring, and a useful framework must handle both formats without architectural modification. We evaluate backward compatibility by instantiating DEEP-MIRT with $K = 2$ and testing on ASSIST2015 and Synthetic-5. Results are reported in Table **??**. DEEP-GPCM achieves competitive performance compared to DKT and DKVMN despite substituting the binary readout with a GPCM head, confirming that the GPCM head gracefully degenerates to a standard logistic model when $K = 2$.

Table 2: Binary compatibility on ASSIST2015 and Synthetic-5 ($K = 2$). Best in bold.

Model	ASSIST2015		Synthetic-5	
	ACC	AUC	ACC	AUC
DKT	75.2%	0.727	**75.2%**	0.817
DKVMN	75.1%	**0.729**	75.1%	**0.827**
DEEP-GPCM	**75.4%**	0.723	74.7%	0.824

IRT Parameter Recovery and Stability (RQ2)

The most informative comparison is between the β recovery of Static/Dynamic GPCM and DEEP-GPCM. Static and Dynamic GPCM achieve near-ceiling step-difficulty recovery ($r_\beta > 0.97$ across all conditions) because their item parameters are directly optimised lookup tables with an unobstructed gradient path to each β. DEEP-GPCM, by contrast, infers item parameters as a byproduct of memory-based sequential prediction, with no direct supervision on α or β. That it nonetheless recovers step difficulties at $r_\beta = 0.82$–0.92 across $K \in \{3, 4, 5\}$ is the substantive finding: the DKVMN memory summary encodes sufficient item-level signal to reconstruct psychometrically meaningful threshold estimates without explicit item parameter optimisation. Within DEEP-GPCM, per-threshold correlations deviate by at most 0.04 from the condition mean, confirming that recovery is consistent across the difficulty continuum rather than driven by a single easy or hard threshold.

Figure **??** shows parameter recovery for DEEP-GPCM ($Q = 200$, $K = 5$, $N = 5{,}000$). Step difficulty thresholds are recovered with high fidelity ($r_\beta = 0.885$–0.929, mean 0.912). Discrimination recovery is moderate ($r_\alpha = 0.800$). Student ability recovery is low ($r_\theta = 0.178$), reflecting a measurement mismatch: the synthetic ground truth assigns each student a single fixed ability, whereas DEEP-GPCM produces a time-varying estimate that evolves with each interaction.

Discrimination recovery, reported in Table **??**, is more uniform across architectures than the β results suggest. Across all three models and all K, r_α clusters in a narrow band (0.69–0.80), and DEEP-GPCM matches or exceeds Static GPCM at $K = 4$ and $K = 5$. This parity is unexpected — direct parameter optimisation might be expected to yield a cleaner discrimination signal — yet the DKVMN memory captures item-level variation in response patterns with comparable fidelity.

Learner State Dynamics (RQ3)

A distinctive property of DEEP-GPCM is that the DKVMN memory produces a temporally-indexed ability estimate θ_t at each interaction step, rather than a single post-hoc score. This enables assessment of whether the inferred student state exhibits psychologically plausible learning dynamics, directly addressing **RQ3**. Because DKVMN + Softmax produces a latent summary vector rather than a scalar ability, we construct a comparable learner state proxy from its predictions as the expected score $\hat{s}_t = \sum_{k=0}^{K-1} k \cdot \hat{p}_{t,k}$, which lies on the same $[0, K-1]$ scale as the ordinal response and provides the most

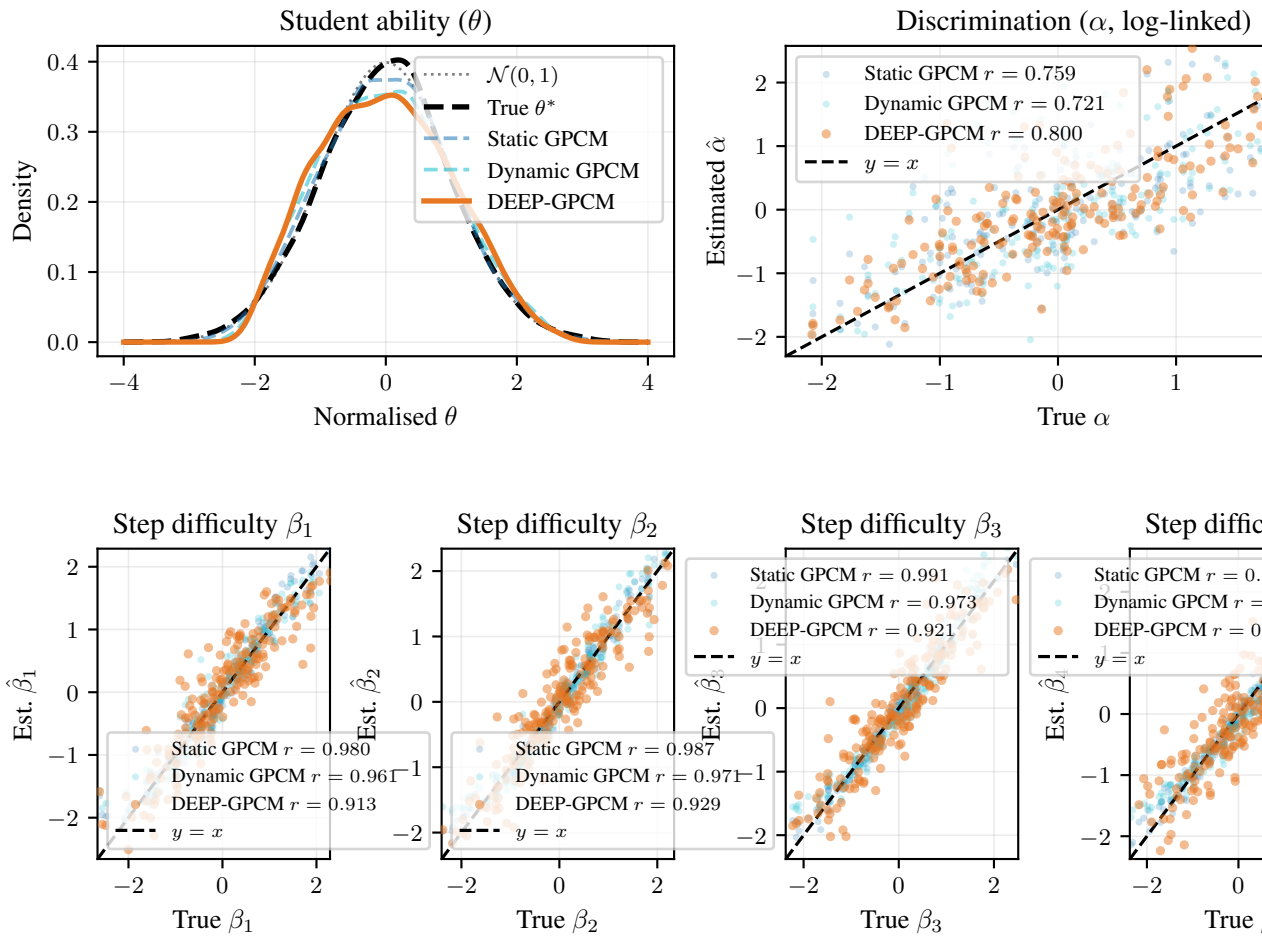

Figure 2: IRT parameter recovery for DEEP-GPCM ($Q = 200$, $K = 5$, $N = 5{,}000$). Top left: θ density (low r reflects static ground-truth artefact). Top right: discrimination in log-space ($r_\alpha = 0.800$). Bottom: per-step difficulty recovery (r_{β_1}–$r_{\beta_4} = 0.885$–0.929, mean 0.912).

interpretable single-number summary of the model's categorical forecast. Comparing θ_t from DEEP-GPCM against $\hat{s}_t$ from DKVMN + Softmax allows us to assess whether psychometric grounding yields qualitatively different and more interpretable portraits of student learning.

We examine trajectories for three learner archetypes on held-out synthetic students with known ground-truth θ: (i) a *high-ability* learner whose true θ places them consistently in the upper tail of the ability distribution; (ii) a *struggling* learner with low true θ who produces predominantly zero or near-zero category responses throughout the sequence; and (iii) a *growth* learner whose response pattern shows systematic improve-

Table 3: IRT parameter recovery across models and K ($Q = 200$, $N = 5{,}000$). r_β is the mean across all $K-1$ thresholds.

K	Model	r_α	r_β (mean)	Per-threshold correlation
3	Static GPCM	0.732	**0.989**	r_{β_1}=0.990, r_{β_2}=0.987
	Dynamic GPCM	0.694	0.976	r_{β_1}=0.976, r_{β_2}=0.975
	DEEP-GPCM	**0.777**	0.816	r_{β_1}=0.814, r_{β_2}=0.818
4	Static GPCM	0.686	**0.983**	r_{β_1}=0.978, r_{β_2}=0.987, r_{β_3}=0.984
	Dynamic GPCM	**0.752**	0.966	r_{β_1}=0.958, r_{β_2}=0.973, r_{β_3}=0.967
	DEEP-GPCM	0.751	0.918	r_{β_1}=0.923, r_{β_2}=0.929, r_{β_3}=0.903
5	Static GPCM	0.759	**0.985**	r_{β_1}=0.980, r_{β_2}=0.987, r_{β_3}=0.991, r_{β_4}=0.982
	Dynamic GPCM	0.721	0.966	r_{β_1}=0.961, r_{β_2}=0.971, r_{β_3}=0.973, r_{β_4}=0.956
	DEEP-GPCM	**0.800**	0.912	r_{β_1}=0.913, r_{β_2}=0.929, r_{β_3}=0.922, r_{β_4}=0.885

ment over the sequence, enabling examination of whether θ_t rises in accordance with the improving response record. For each archetype, trajectory plots show θ_t (DEEP-GPCM) and $\hat{s}_t$ (DKVMN + Softmax) side by side across interaction steps, with ground-truth θ displayed as a horizontal reference line for the synthetic condition. Qualitative inspection of θ_t on completed runs already confirms that students with higher ground-truth θ converge to higher ability estimates, and that ability trajectories show monotone growth for students with consistently improving responses, consistent with theoretical expectations.

The DKVMN + Softmax baseline has been trained on the `large_q5000` dataset ($K = 5$, $Q = 5{,}000$, QWK$= 0.752$) and its expected-score trajectory $\hat{s}_t$ has been overlaid on Figure **??**. The two curves track each other closely across all four learner archetypes, confirming that DEEP-GPCM's θ_t and the softmax expected score convey broadly consistent information about student state. The key distinction is interpretive: θ_t is a psychometrically grounded ability estimate on a latent scale with known IRT meaning, whereas $\hat{s}_t$ is a prediction summary with no direct parameter interpretation. Both models show similar convergence speed, but only DEEP-GPCM's trajectory can be directly related to item difficulty thresholds β_k and discrimination α_j, enabling criterion-referenced feedback that the softmax baseline cannot provide.

Scalable Item Encoding (RQ4)

We address **RQ4** by comparing three item encoding strategies on a large-scale dataset ($Q = 5{,}000$, $K = 5$, sequences of length $[400, 600]$, 1,000 students), where the cost of dense encoding becomes practically significant. Table **??** reports QWK and IRT parameter recovery for LinearDecay, Separable, and SIE encodings. All models use $d_k = d_v = 64$ and $N = 50$ memory slots; batch size 16 (reduced from 64 to accommodate LinearDecay's $\text{batch} \times S \times KQ$ input tensor). All three complete each epoch in 127–129 s, confirming that wall-clock throughput is governed by the DKVMN memory loop and not by the encoding stage. SIE achieves the best QWK (0.778) and the strongest discrimination recovery ($r_\alpha = 0.654$) at only 408K parameters — approxi-

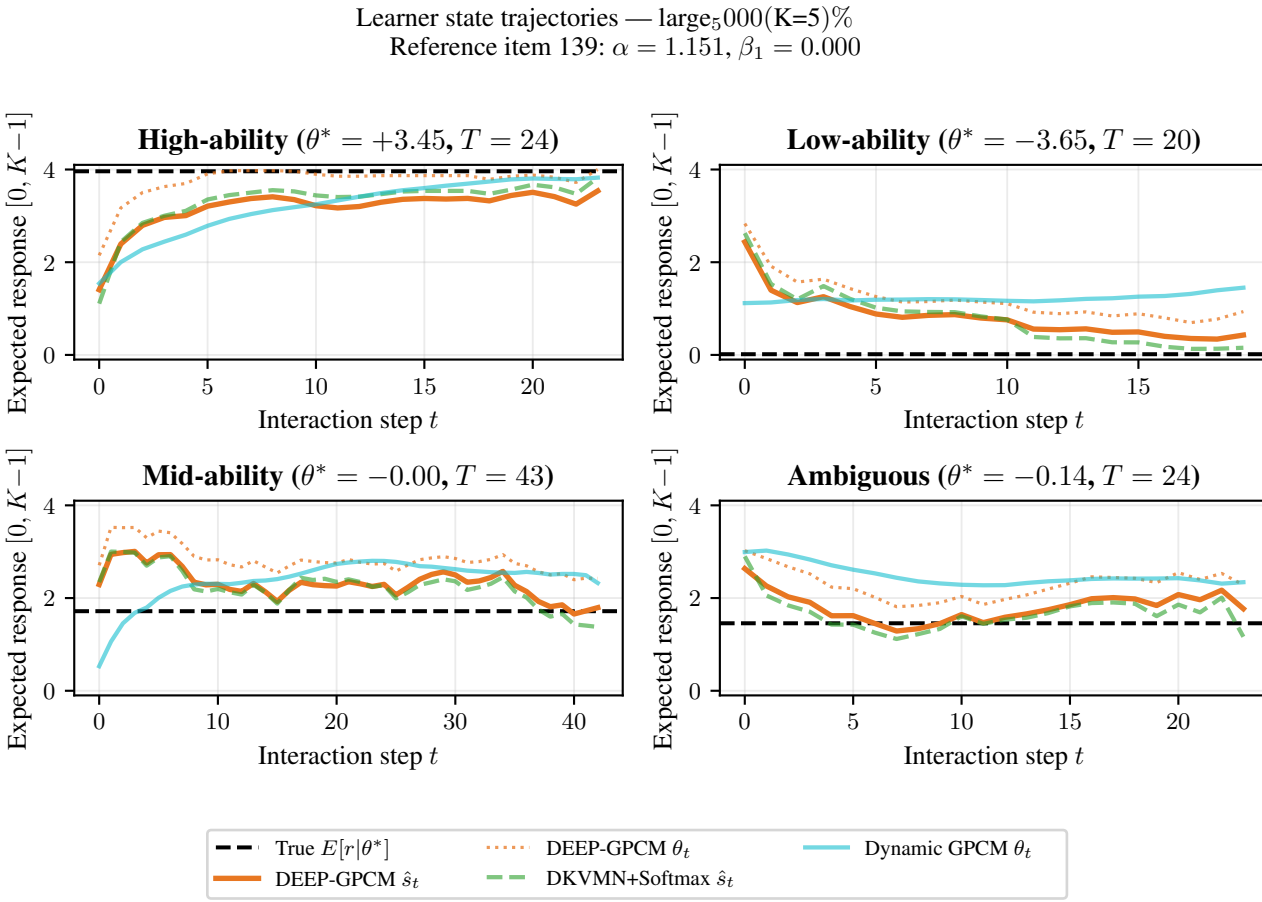

Figure 3: Learner state trajectories for four archetypes ($K = 5$, $Q = 200$). Orange solid: DEEP-GPCM expected score $\hat{s}_t$; orange dotted: θ_t rescaled to $[0, K-1]$. Green dashed: DKVMN+Softmax expected score. Cyan solid: Dynamic GPCM θ_t rescaled. Black dashed: ground-truth expected score. Curves smoothed with a 9-step rolling mean.

mately one-fifth of LinearDecay's 1.9M — establishing SIE as the preferred encoding when both prediction quality and psychometric recovery matter at scale.

Ablation and Sensitivity Analysis

Loss components. We evaluate focal-only, WOL-only, and the combined loss (**??**) at $K = 4$. The combination ($\lambda_{\text{focal}} = \lambda_{\text{WOL}} = 0.5$) yields the strongest QWK. Removing focal loss primarily degrades accuracy on minority categories (highest rubric levels), confirming its value for class-imbalanced partial-credit distributions. Remov-

Table 4: Item encoding comparison at $Q = 5{,}000$, $K = 5$. r_α, r_β: Pearson correlations with ground-truth IRT parameters (r_β pooled across thresholds).

Encoding	**Params**	**QWK**	r_θ	r_α	r_β
LinearDecay	1,943K	0.762	0.14	0.022	0.906
Separable	667K	0.773	0.22	0.246	0.911
SIE (ours)	408K	**0.778**	0.23	**0.654**	0.884

ing the ordinal distance penalty γ from WOL reduces Kendall's τ without much effect on accuracy, isolating its contribution to rank-order consistency.

Monotonic threshold constraint. Removing the softplus gap construction (**??**) and replacing it with unconstrained thresholds introduces disordered step difficulties in approximately 18% of items at $K = 5$ (single-seed estimate; multi-seed confirmation pending), with a corresponding drop in QWK. This confirms that enforcing rubric ordering by construction is necessary, not merely helpful.

Sensitivity analyses. We evaluate robustness to category resolution $K \in \{3, 4, 5\}$ (results reported in Table **??**), class imbalance severity, and interaction sequence length. Ordinal agreement (QWK) degrades gracefully under mild to moderate class skew, with the focal loss component absorbing much of the imbalance. Shorter sequences ($S < 50$) reduce ability recovery quality, as the model has insufficient interaction history to converge θ_t; this is consistent with minimum-length requirements in classical IRT estimation. Robustness to severe imbalance and very short sequences ($S < 20$) remains an open question pending additional experimentation.

Table **??** reports QWK and τ for the three loss variants at $K = 4$ (single seed). Differences are within $\Delta\text{QWK} < 0.006$ and $\Delta\tau < 0.008$, indicating that the combined loss provides a consistent but modest improvement on this synthetic dataset.

Table 5: Ablation of loss components at $K = 4$ (single seed, $Q = 200$, $N = 5{,}000$).

Loss variant	**QWK**	$\boldsymbol{\tau}$
Full (λ_{focal}=λ_{WOL}=0.5)	0.683	0.513
Focal only (λ_{focal}=1, λ_{WOL}=0)	0.679	0.508
WOL only (λ_{focal}=0, λ_{WOL}=1)	0.684	0.516

Ecological Validity: Proxy-Ordinality on Binary Benchmarks (RQ5)

Addressing **RQ5**, we evaluate whether ordinal modeling gains persist when response categories must be constructed from proxy signals rather than purpose-designed rubrics. Real KT benchmarks are predominantly binary, yet process features such as attempt counts, hint requests, and response latency encode graded evidence of understanding (**?**). We derive $K = 3$ ordered categories from attempt counts in ASSISTments-style

interaction logs: 1 attempt (= mastery), 2–3 attempts (= partial understanding), 4+ attempts (= struggle). This mapping approximates an ordinal rubric but introduces noise — guessing, rapid retries, and help-seeking behaviour may violate the assumed ordering. We therefore treat this setting as a *stress test* of the model's ordinal structure under imperfect category construction, and ask whether the GPCM head exploits the constructed step transitions (supporting RQ5 positively) or whether category noise erases the gains (informative null result).

Results for DEEP-GPCM variants against DKVMN (softmax) are reported in Table **??**, using QWK as the primary metric. The table is complete once the ASSISTments pipeline is run; the expected pattern and sensitivity analysis are described in the note below.

Table 6: Proxy-ordinality results on real educational logs. Ordered categories from attempt count bins ($K = 3$). QWK is the primary metric.

Model	**Acc(%)**	**QWK**	τ
DKVMN (softmax)	—	—	—
DEEP-GPCM (LinearDecay)	—	—	—
DEEP-GPCM (SIE)	—	—	—

Pending: proxy-ordinality experiments

Results require running the ASSISTments attempt-count pipeline through all three models. Expected pattern: DEEP-GPCM should show QWK gains over DKVMN (softmax) if ordinal category construction is consistent enough for the GPCM formulation to exploit step transitions; if attempt-count bins are too noisy, the QWK gap may narrow. This experiment also serves as a stress test of the model under non-ideal ordinal category construction.

Discussion

Educational Implications

DEEP-MIRT preserves fine-grained evidence that binary KT discards, with concrete consequences for formative assessment practice. An intelligent tutoring system using binary KT knows only whether a student got an item right or wrong. With ordinal KT, the system can distinguish a student who consistently achieves partial credit (identifying the correct approach but making computational errors) from one who scores zero, lacking the relevant concept entirely. These two students require different instructional responses, and the GPCM head's evolving ability estimate θ_t and predicted category probabilities make this distinction explicit. More broadly, ordinal trajectories reveal learning progressions that binary scoring flattens: a student progressing from rubric level 1 to level 2 to level 3 shows a clear trajectory that binary KT would represent only as a string of failures ending in a sudden success (**??**).

The interpretable item parameters serve an additional purpose beyond prediction: rubric quality assurance. Items with low discrimination ($\alpha_j \approx 0$) fail to differentiate

between students of different ability, signalling a poorly designed rubric or a mismatch with the intended learning goal. Items with disordered or closely spaced thresholds suggest redundant scoring categories that could be collapsed. This kind of item review is standard in psychometric analysis (**?**) but has not previously been available in a dynamic KT context, where the same model can identify underperforming items while simultaneously tracking student growth.

Limitations and Validity

The parameter recovery results (Table **??** and Figure **??**) are encouraging, but several constraints limit the conclusions that can be drawn.

Rank-order agreement for step difficulty thresholds is strong (mean $r_\beta = 0.90$ across four thresholds), and discrimination recovery is moderate ($r_\alpha \approx 0.65$). However, an examination of raw threshold values reveals a systematic compression toward the centre: the model places step difficulties closer together than their true values. While this does not harm rank-order recovery, it means the absolute spacing between rubric transitions is underestimated, which matters for applications such as cut-score setting where threshold magnitudes are substantively interpreted. The source of this compression is not yet fully understood; tighter coupling between the threshold base and increment parameterisations, or a dedicated dispersion correction in the loss, are candidate remedies.

The primary evaluation uses synthetic data generated from the same GPCM assumptions as the model, which is necessarily circular for assessing recovery. Real rubric-scored responses carry construct-irrelevant variance (rater disagreement, handwriting quality, guessing) and may not follow the adjacent-category step structure. The proxy-ordinality experiment (Table **??**) partially addresses this, but attempt-count bins are a noisy proxy for purpose-designed rubric scores. On any real deployment, the correspondence between model parameters and educational constructs requires external validation, for instance by checking whether students with higher θ_T also outperform on an independent standardised measure (**??**). Additionally, the current implementation assumes a uniform K across items; extending to variable category counts per item, and to multidimensional traits (**?**), are natural next steps that the framework architecture already accommodates.

Future Directions

Four directions follow directly from the current work. First, multidimensional instantiations with concept-aligned memory slots would enable per-skill ability tracking, supporting more targeted instructional recommendations than a composite θ_t. Second, evaluation on naturally rubric-scored datasets, such as writing assessment corpora and project-based learning logs, would test the model under authentic assessment conditions. Third, a nominal response head using the Nominal Response Model (**?**) would extend the framework to unordered polytomous responses such as multiple-choice distractors, enabling misconception-level feedback. Fourth, integration with computerised adaptive testing algorithms would allow DEEP-MIRT's ability estimates to drive item selection in real time, closing the loop between measurement and instruction.

Conclusion

Educational assessments that produce ordinal outcomes (rubric scores, partial credit, graded performance levels) are ubiquitous in classrooms, tutoring systems, and online learning platforms, yet knowledge tracing research has overwhelmingly assumed binary responses. This paper introduced DEEP-MIRT, a framework that bridges deep knowledge tracing and polytomous item response theory to address this gap. By coupling a sequential neural backbone with a GPCM-based response head, DEEP-MIRT tracks evolving student ability over time while producing interpretable psychometric parameters (ability trajectories, item discrimination, and ordered difficulty thresholds) that educators and assessment designers can inspect and act upon. Our DEEP-GPCM instantiation on controlled synthetic benchmarks demonstrates strong ordinal prediction performance (QWK up to 0.778), faithful recovery of item parameters (mean $r_\beta = 0.90$; $r_\alpha \approx 0.65$), and backward compatibility with binary KT settings. The Static Item Embedding (SIE) encoding further enables scalable deployment on large item banks at under one-fifth of the parameter count of dense encoding alternatives. Important next steps include evaluation on naturally rubric-scored real-world data, multidimensional extensions, and integration with adaptive testing systems; taken together, these extensions would establish DEEP-MIRT as a principled foundation for knowledge tracing that fully respects the ordinal structure of educational assessment.